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import sqlite3

def create_database():
    conn = sqlite3.connect("shop.db")
    cursor = conn.cursor()
    cursor.execute('''CREATE TABLE IF NOT EXISTS products (
                        id INTEGER PRIMARY KEY AUTOINCREMENT,
                        name TEXT NOT NULL,
                        price REAL NOT NULL,
                        stock INTEGER NOT NULL
                    )''')
    conn.commit()
    conn.close()

def insert_products():
    conn = sqlite3.connect("shop.db")
    cursor = conn.cursor()

    # Check if products exist to avoid duplicate insertions
    cursor.execute("SELECT COUNT(*) FROM products")
    if cursor.fetchone()[0] == 0:
        products = [
            ("Laptop", 750.99, 10),
            ("Mouse", 25.50, 50),
            ("Keyboard", 45.75, 30),
            ("Monitor", 180.25, 15),
            ("Headphones", 60.00, 20)
        ]
        cursor.executemany("INSERT INTO products (name, price, stock) VALUES (?, ?, ?)", products)
        conn.commit()

    conn.close()

def display_products():
    conn = sqlite3.connect("shop.db")
    cursor = conn.cursor()
    cursor.execute("SELECT * FROM products")
    products = cursor.fetchall()
    conn.close()

    print("\nAvailable Products:")
    for product in products:
        print(f"ID: {product[0]}, Name: {product[1]}, Price: ${product[2]:.2f}, Stock: {product[3]}")
    print("-" * 40)

def add_to_cart(product_id):
    conn = sqlite3.connect("shop.db")
    cursor = conn.cursor()

    # Check if the product exists and stock is available
    cursor.execute("SELECT stock FROM products WHERE id = ?", (product_id,))
    result = cursor.fetchone()

    if result:
        stock = result[0]
        if stock > 0:
            cursor.execute("UPDATE products SET stock = stock - 1 WHERE id = ?", (product_id,))
            conn.commit()
            print(f"Product ID {product_id} added to cart.")
        else:
            print("Product is out of stock.")
    else:
        print("Invalid Product ID.")

    conn.close()

def remove_from_cart(product_id):
    conn = sqlite3.connect("shop.db")
    cursor = conn.cursor()

    # Ensure we don't remove more than the original stock limit
    cursor.execute("UPDATE products SET stock = stock + 1 WHERE id = ?", (product_id,))
    conn.commit()
    conn.close()
    print(f"Product ID {product_id} removed from cart.")

def total_bill():
    conn = sqlite3.connect("shop.db")
    cursor = conn.cursor()

    # Retrieve all products where stock has been reduced (items added to cart)

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cursor.execute("""
    SELECT name, price, (10 - stock) AS quantity
    FROM products WHERE (10 - stock) > 0
""")

cart_items = cursor.fetchall()
conn.close()

total = 0
print("\nCart Summary:")
for item in cart_items:
    if item[2] > 0: # Ensure quantity is valid
        item_total = item[1] * item[2]
        total += item_total
        print(f"{item[0]} - ${item[1]:.2f} x {item[2]} = ${item_total:.2f}")

print(f"Total Bill: ${total:.2f}")
print("-" * 40)

# Create database and insert products
create_database()
insert_products()
display_products()

# Add products to cart
add_to_cart(1) # Laptop
add_to_cart(3) # Keyboard

# Display bill
total_bill()

# Remove an item and show the updated bill
remove_from_cart(1)
total_bill()

# Show updated product list
display_products()

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Available Products:
ID: 1, Name: Laptop, Price: $750.99, Stock: 9
ID: 2, Name: Mouse, Price: $25.50, Stock: 50
ID: 3, Name: Keyboard, Price: $45.75, Stock: 29
ID: 4, Name: Monitor, Price: $180.25, Stock: 15
ID: 5, Name: Headphones, Price: $60.00, Stock: 20
-----
Product ID 1 added to cart.
Product ID 3 added to cart.

Cart Summary:
Laptop - $750.99 x 2 = $1501.98
Total Bill: $1501.98
-----
Product ID 1 removed from cart.

Cart Summary:
Laptop - $750.99 x 1 = $750.99
Total Bill: $750.99
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Available Products:
ID: 1, Name: Laptop, Price: $750.99, Stock: 9
ID: 2, Name: Mouse, Price: $25.50, Stock: 50
ID: 3, Name: Keyboard, Price: $45.75, Stock: 28
ID: 4, Name: Monitor, Price: $180.25, Stock: 15
ID: 5, Name: Headphones, Price: $60.00, Stock: 20
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